

Sungyeon Kook

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Research Interests

Quantum Error Correction, Fault-Tolerant Quantum Computing, Quantum Information Theory, Quantum Algorithms, Error-Correcting Codes

Education

Korea University M.S. (Master of Science) in Electrical Engineering	SEOUL, KOREA Mar. 2024 – Feb. 2026
Korea University B.S. (Bachelor of Science) in Electrical Engineering	SEOUL, KOREA Mar. 2020 – Feb. 2024

Research Experience

Korea Institute of Science and Technology Information (KISTI) <i>Post-Master Researcher</i> , Quantum Network Research Center (Advisor: Dr. Ilkwon Sohn)	DAEJEON, KOREA Mar. 2026 – Present
Korea University <i>Graduate Research Assistant</i> , Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo)	SEOUL, KOREA Mar. 2024 – Feb. 2026
Korea Research Institute of Standards and Science (KRISS) <i>Visiting Student Researcher</i> , Center for Superconducting Quantum Computing System (Advisor: Dr. Hwan-Seop Yeo)	DAEJEON, KOREA Aug. 2024 – Aug. 2025
Korea University <i>Undergraduate Research Assistant</i> , Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo) <i>Undergraduate Research Assistant</i> , Quantum Control Laboratory (Advisor: Prof. Mahn-Soo Choi)	SEOUL, KOREA Jan. 2023 – Feb. 2024 Jul. 2023 – Aug. 2023

Publications

- Changyeol Lee[†], **Sungyeon Kook**[†], Wooyeong Song, Kwangil Bae, and Ilkwon Sohn, “Quantum Crandall reduction for pseudo-Mersenne moduli: structure-aware circuits and surface-code resource analysis,” manuscript in preparation for submission to [Quantum Science and Technology](#). ([†] Equal contribution.)
- Wooyeong Song, **Sungyeon Kook**, Wonhyuk Lee, and Ilkwon Sohn, “Asymmetry-aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client,” under review at [IEEE Network](#). arXiv:2607.11238.
- Sungyeon Kook**, Yujin Kang, Ilkwon Sohn, and Jun Heo, “Low Spatial Cost CCZ Magic State Factory,” under review at [Quantum Information Processing](#). arXiv:2606.24170.
- Sungyeon Kook**, Yujin Kang, and Jun Heo, “Limited Connectivity Quantum Architectures: Enhancing Surface Code Implementation Through Routing Qubits,” in [Proceedings of the 15th International Conference on Information and Communication Technology Convergence \(ICTC\)](#), 2024. DOI: 10.1109/ICTC63764.2024.10826938.
- Yujin Kang, Youshin Chung, Huidan Zheng, **Sungyeon Kook**, and Jun Heo, “Magic State Distillation with Reduced Time Cost,” in [Proceedings of the IEEE International Conference on Quantum Computing and Engineering \(QCE\)](#), 2024. DOI: 10.1109/QCE60285.2024.00142.

Conferences

Poster Presenter

- 2026 Annual Meeting of the Quantum Information Society of Korea 2026.02
- International Conference on Information and Communication Technology Convergence (ICTC) 2025.10
- **Quantum Error Correction Conference (QEC 2025)** 2025.08

Oral Presenter

- Joint Conference on Communications and Information (JCCI) 2025.04
- International Conference on Information and Communication Technology Convergence (ICTC) 2024.10
- KICS Conferences (Multiple Oral Presentations) 2023–2025

Co-author

- **Asian Quantum Information Science Conference (AQIS 2026)** 2026.08
- Optics and Photonics Congress 2026 (OPC 2026) 2026.07
- **IEEE International Conference on Quantum Computing and Engineering (QCE)** 2024.09

Research Projects

** Journal published or under review*

Low-Spatial-Cost CCZ Magic-State Factory*

2025–Present

- Converted the $[[8, 3, 2]]$ CCZ distillation circuit into **eight Z-basis $\pi/8$ Pauli-product joint measurements**, reducing the factory from **12 to 4 persistent logical patches**.
- Eliminated **eight redundant $|0\rangle$ ancilla qubits** while preserving logical functionality and single-fault detection.
- Designed a surface-code CCZ factory using **four-qubit twisted ZY measurements**, reducing spatial cost by **71.4%** and space-time volume by **37.7%**.

Surface-Code Decoder and Code Switching*

2024–Present

- Built a **Stim–PyMatching decoding pipeline** for rotated surface codes, evaluating logical failure rates across code distances and physical-error rates.
- Developed an **erasure-aware MWPM decoder** by dynamically reweighting matching-graph edges using known erasure locations.
- Proposed a fault-tolerant **Bacon–Shor-to-surface-code switching protocol** for heterogeneous quantum computation.

Architecture-Aware Quantum Circuit Mapping and Scheduling

2024–2025

- Designed heavy-hex surface-code mapping strategies to minimize SWAP overhead while balancing flag-qubit and hardware resources.
- Developed a **QODG-based scheduling framework** for checkerboard architectures by modeling gate dependencies, resource conflicts, and conditional execution latency.
- Implemented a Greedy DAG Qiskit transpiler pass, reducing average circuit depth from **73 to 61.4**.

Fault-Tolerant Resource Analysis for Quantum Modular Reduction*

2025–Present

- Developed a **surface-code resource-estimation framework** incorporating code distance, logical-error requirements, magic-state factories, decoder throughput, and backlog.
- Formulated a **fault-tolerant runtime model** incorporating T-depth, feed-forward latency, and decoder performance.
- Reduced estimated execution time from **53.77 ms to 30.05 ms** under Sparse Blossom decoding.

Experimental Superconducting Quantum Computing

KRISS, 2024–2025

- Developed Python-based experimental-control software for superconducting-qubit experiments using Zurich Instruments hardware.
- Performed **qubit calibration and characterization**, including spectroscopy, randomized benchmarking, DRAG calibration, and cross-resonance gates.
- Investigated heavy-hex architectures qubit routing, flag-qubit syndrome extraction, and connectivity-aware fault-tolerant circuits.

Variational Quantum Algorithms

2023

- Implemented VQE, QAOA, and quantum-kernel algorithms using Qiskit and PennyLane.

Technical Skills

Programming: Python, C++, MATLAB

Quantum Software: Qiskit, PennyLane, Stim, PyMatching

Experimental Platforms: Zurich Instruments, Superconducting Quantum Computing

Professional Activities

IBM Quantum

Sep. 2025 – Present

Qiskit Advocate

Quantum Information Club at Korea University (QUICK)

Jan. 2023 – Dec. 2023

Founder and Treasurer

- Co-founded the first undergraduate quantum information club at Korea University.
- Managed club administration and event organization.
- Organized the first IBM Quantum Fall Fest hosted at Korea University.

Teaching & Honors

Outstanding Academic Achievement, Korea University

Spring 2023

Teaching Assistant, Data Science and Artificial Intelligence

Fall 2024

Provided programming and data analysis instruction for undergraduate students.

Teaching Assistant Scholarship, Korea University

Fall 2024